

蔡秉叡 Ray Tsai

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Research Interests

Multimodal human-computer interaction, with focus on voice-driven AI systems that adapt to users rather than the other way around: own-voice speech coaching, accent-aware L2 pronunciation training, and face/voice/gesture-mediated interfaces. Broader interests include agentic coding workflows, decentralized AI infrastructure (Web4 / OG), and low-resource language preservation through speech technology.

Education

National Taiwan University (NTU), Taipei, Taiwan *Feb 2026 – Present*

M.S., Graduate Institute of Computer Science and Information Engineering (CSIE)

- Member of **HCI Lab** led by [Prof. Mike Y. Chen](#) (mikechen.com) — research on multimodal interaction, emotional voice model and AR/VR interfaces
- Selected coursework (114-2, Spring 2026): Multimodal Human-Computer Interaction (CSIE7641), Natural Language Processing (CSIE5042), Artificial Intelligence (CSIE5400), Big Data Systems (CSIE5322), Operating Systems (CSIE3310), Web Programming (CSIE5210)

National Yang Ming Chiao Tung University (NYCU), Hsinchu, Taiwan

Sep 2021 – Dec 2025

B.S. in Electrophysics

- Cumulative GPA: 4.1 / 4.3 over the final four semesters
- Graduate-level coursework outside the physics core: Machine Learning for Signal Processing (A+), Speech Signal Processing (A+), Deep Learning (A), Artificial Intelligence and Law (A+)

Selected Projects

Mirror — AI Speech Coach [🔗](#)

Apr 2026 – Present

Solo Founder; 2026 NTU AI Builders Challenge

- Built an end-to-end speech-coaching app that records the user's spoken answer, analyses delivery, and replays an *optimized* version in the user's own cloned voice — a direction that few coaching tools take
- Stack: Expo / React Native, Next.js, Supabase, typed-episodic memory store with LUFY retention scoring and LLM-mediated recall, emotional voice clone and per-sentence streaming TTS
- Trilingual practice (Traditional Chinese, English, Japanese); shipped onboarding, voice calibration, mock-interview flows, credits / referral loop, and App Store-compliant account deletion
- **Traction (@mirror_app_ntu [🔗](#), zero ad spend): 502 signups in ~4 weeks with 90.2% onboarding completion, 556 mock-interview sessions** from 191 activated users; single-day peak of **101 signups / 108 DAU** on May 13 from one Threads post — Threads alone accounted for **29.7%** of total acquisition

Instabrain Web4.0 [🔗](#)

Mar 2026 – Present

Decentralized extension of the Mirror voice-AI core

- Re-architected Mirror's persona / voice / memory core onto Web4 infrastructure: Privy embedded wallets, OG Compute Router (TEE-verified inference on OGM-1.0-35B-A3B), OG Storage for memories, and ERC-7857 *iNFT* personas on OG Chain
- Implemented streaming SSE chat with TEE attestation, memory extraction & recall over decentralized storage, and a Hardhat-deployed `InstabrainPersona.sol` contract that ties persona ownership to the user's wallet
- Demonstrates a path for own-voice AI personas in which users — not platforms — hold both the model artifacts and the verifiable compute receipts

Shadow Your Perfect Self

May 2026 – Present

CSIE7641 Multimodal HCI Final Project, NTU

- Designed and is piloting a three-condition shadowing experiment comparing native-stranger, native-accent self-clone, and mild-L1-accent self-clone model voices for Mandarin-L1 English learners
- Hypothesis grounded in cognitive load theory (Sweller), ZPD (Vygotsky), and the voice-confrontation literature: that minimising both timbre and accent distance from the learner may outperform fully native targets

Tinder Face Controller

Mar 2026 – Apr 2026

CSIE7641 Multimodal HCI Mini Project [/github/tinder-face-controller](#)

- Hands-free control of **tinder.com** via webcam: MediaPipe FaceLandmarker + OpenCV **solvePnP** recover yaw/pitch/roll; a Tampermonkey userscript polls a local Python backend at 10 Hz and dispatches synthetic DOM events
- Gestures: nod = like, shake = nope, gaze = photo navigation, mouth open = open profile; optional **Auto Mode** streams frames to Hume AI and learns the user's like / nope emotion vectors after 10 manual swipes
- No patching of the host site; head-pose smoothing via One-Euro filter; sliding-window gesture detection robust to one-direction nods

Royalty Mapping Pro

2025 – Present

Browser-based AI projection-mapping tool [/github/royalty-mapping-pro](#)

- Built a zero-install web projection-mapping app combining classic 4-point corner-pinning / blend modes with Google Veo text-to-video and image-to-video generation for live texture creation
- Dual-window output via the Document Picture-in-Picture API with a BroadcastChannel + LocalStorage fallback so editor and projector stay in millisecond sync; auto-scaling for arbitrary projector resolutions

Research Experience

Research Assistant (Part-time)

Jun 2025 – Mar 2026

CKIP Lab, Institute of Information Science, Academia Sinica

- Worked under Dr. Wei-Yun Ma on multimodal language models and emotion-aware speech generation
- Contributed to the Formosa Speech Workshop Challenge 2025 team: progressive SpecAugment training schedule with dialect-aware conditioning for low-resource Hakka ASR
- Surveyed the Kimi-Audio architecture for joint emotion recognition and emotion-controllable speech synthesis

Research Assistant

Mar 2023 – Jul 2025

Institute of Learning Sciences & Technologies, National Tsing Hua University

- Co-developed a multimodal AI picture-book platform combining voice synthesis and image generation for language learning and cognitive therapy
- Delivered hands-on workshops (12 hours / semester) introducing graduate students to generative AI, n8n / LangChain workflows, and agentic automation

Leadership & Professional Experience

Public Relations Officer

2022 – 2023

Department of Electrophysics Student Association, NYCU

- Hosted 9 career seminars with 700+ total attendees; secured speakers from major semiconductor and photonics firms

Research Intern, ITRI | IT Intern, DuPont Taiwan

Oct 2022 – Jun 2023

- ITRI: Python web-scraping tools for automated data collection. DuPont: corporate IT workshops and network maintenance